Do Now · retrieval

Normalisation to 3NF and indexing

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a transitive dependency?

Q2

What does an index help with?

Q3

What does 1NF require?

Q4

Simplify (A∧B)∨(A∧¬B), stating the Boolean law.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — 1NF requires atomic values and no repeating groups in a record; 2NF removes dependencies on only part of a composite candidate key.

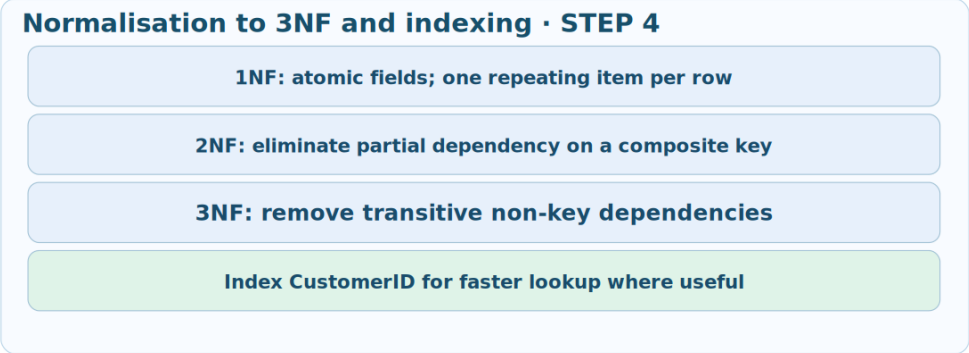
Core idea 2

Apply — 3NF eliminates inappropriate transitive dependencies among non-key facts, reducing repeated customer names across orders.

Core idea 3

Evaluate — An index accelerates some lookups but uses storage and may slow insert/update operations. Normalisation and indexing solve different problems.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

1NF — Atomic values/no repeating groups.

2NF — No partial dependency on part of composite key.

3NF — No non-key transitive dependency violating key dependency.

index — Search structure over columns.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

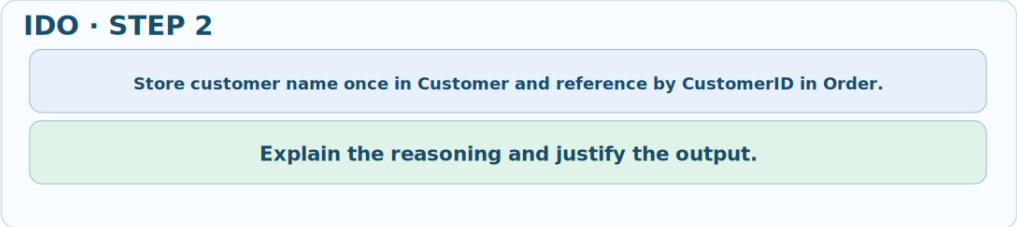
Problem

A table Order(OrderID,CustomerID,CustomerName) repeats a name. Show a 3NF decomposition.

Follow the worked method

1. Predict first — Store customer name once in Customer and reference by CustomerID in Order.
2. Model answer — Explain the reasoning and justify the output.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
ProductName depends on ProductID rather than the whole composite sales-line identifier, so separate it.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

State 1NF,2NF,3NF in one sentence each. [6]

Question 2

Order(OrderID, CustomerID, CustomerName). Explain the redundancy and split into two tables. [6]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why an index on StudentName can speed SELECT but also incur costs. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Normalising to 3NF means the database never needs indexes.

Use key vocabulary from memory

1. Explain 1NF in your own words.

2. Explain 2NF in your own words.

3. Explain 3NF in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.